

Pearson BTEC Level 3 National Extended Diploma in Engineering.

Unit Identification: Unit 19: Electronic Devices and Circuits.

Assessment Component: Learning Aim C - Improving Personal Performance and Lessons Learnt.

Assignment Title: Assignment C: Improving Personal Performance and Lessons Learnt.

Learner Name: Daniel Boateng.

Learner Number: 83789.

Submission Date: 24th June, 2026.

Report Title

Assignment C: Improving Personal Performance and Lessons Learnt.

Unit 19: Electronic Devices and Circuits.

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This report evaluates the knowledge, skills, techniques and professional behaviours developed throughout Assignments A and B for Unit 19: Electronic Devices and Circuits. The report reflects on practical investigations involving analogue and digital electronic circuits, including rectifiers, voltage regulators, transistor amplifiers, operational amplifiers, combinational logic systems, shift registers and ripple counters. It critically evaluates personal performance, health and safety practices, time management, communication skills and future development opportunities while demonstrating how these experiences support progression into higher education and professional engineering careers.

Assignment C: Improving Personal Performance and Lessons Learnt.

Introduction.

Throughout Assignments A and B, I investigated a wide range of analogue and digital electronic devices and circuits using theoretical calculations, simulation software and practical prototype construction. The analogue investigations included PN junction diodes, half-wave and full-wave rectifiers, smoothing capacitors, Zener diode voltage regulators, transistor amplifiers, MOSFETs and operational amplifier circuits. The digital investigations included combinational logic systems, Boolean algebra, Karnaugh map simplification, NAND-only logic implementation, shift registers and asynchronous ripple counters.

The combination of theoretical calculations, Multisim simulations, CircuitVerse modelling and practical prototype testing significantly improved my understanding of electronic engineering principles. It also allowed me to develop important technical, analytical and professional skills while working safely within laboratory environments.

1. Review and Reflection on Practical Activities.

Knowledge Gained from Analogue Electronics.

During Assignment A, I developed a comprehensive understanding of PN junction theory and the operation of semiconductor devices.

The investigation into rectifier diodes demonstrated how alternating current can be converted into direct current through half-wave and full-wave rectification. The half-wave rectifier illustrated the limitations of using a single diode, particularly the large ripple voltage and poor efficiency. In contrast, the

full-wave bridge rectifier utilised both halves of the AC waveform, producing a smoother DC output and significantly improving efficiency.

The ripple voltage investigation further enhanced my understanding of power supply design. By comparing 100 μ F, 220 μ F, 470 μ F and 1000 μ F smoothing capacitors, I observed how increasing capacitance reduced ripple voltage. This confirmed the theoretical relationship:

$$R.V = I / fC$$

The comparison of calculated, simulated and measured values demonstrated that practical circuits are influenced by component tolerances and non-ideal characteristics.

The Zener diode investigation provided valuable insight into voltage regulation. I learned how Zener diodes operate in reverse breakdown mode to maintain a stable output voltage despite variations in input supply voltage. This demonstrated the importance of regulation circuits within electronic power supplies.

My understanding of transistor devices also improved significantly. I gained insight into the differences between current-controlled BJTs and voltage-controlled MOSFETs. Through the common emitter amplifier investigation, I learned the importance of correct biasing to achieve linear amplification without distortion. The frequency response testing demonstrated how gain decreases at higher frequencies due to the gain-bandwidth limitations of the amplifier.

The operational amplifier investigations introduced both inverting and non-inverting amplifier configurations. I learned how external resistor values determine voltage gain and how op-amps can provide predictable and stable amplification.

Knowledge Gained from Digital Electronics.

Assignment B strengthened my understanding of both combinational and sequential logic systems.

The alarm system investigation required the creation of truth tables and Boolean expressions. I learned how real-world sensor conditions can be represented as binary logic states and processed using digital circuits.

Using Karnaugh maps allowed me to simplify Boolean expressions and reduce the number of logic gates required. This demonstrated how optimisation improves reliability, reduces power consumption and lowers component count.

The implementation of De Morgan's Theorem showed how any logic circuit can be converted into a NAND-only design. Since NAND gates are universal gates, this technique is widely used within commercial digital systems.

The shift register investigation developed my understanding of sequential logic. I learned how D-type flip-flops (74HC74) store and transfer information using clock pulses. The LED progression demonstrated how data moves through successive stages of the register.

The asynchronous ripple counter investigation introduced binary counting and frequency division. Using JK flip-flops (4027B), I observed how the counter progressed through the complete sequence from 000 to 111. This demonstrated how counters are used in digital clocks, timing circuits and industrial automation systems.

I also developed an understanding of the differences between synchronous and asynchronous sequential circuits, particularly the effects of propagation delay in ripple counters.

Skills Developed.

Software and Simulation Skills.

Throughout the assignments I became proficient in using both Multisim and CircuitVerse.

These tools enabled me to:

- Design and simulate analogue circuits.
- Construct digital logic systems.
- Analyse waveform behaviour.
- Validate theoretical calculations.
- Identify design errors before practical implementation.

The ability to simulate circuits before hardware construction improved efficiency and reduced troubleshooting time.

Practical Prototyping Skills.

Practical circuit construction significantly improved my engineering skills.

I developed confidence in:

- Breadboard assembly.
- Component identification.
- Datasheet interpretation.

- Oscilloscope operation.
- Multimeter measurements.
- Function generator configuration.
- Hardware troubleshooting.

Constructing physical circuits highlighted the importance of correct component orientation, accurate wiring and systematic testing.

Analytical and Problem-Solving Skills.

The investigations required continuous comparison between:

- Theoretical calculations.
- Simulation results.
- Practical measurements.

This strengthened my ability to identify discrepancies, determine root causes and evaluate circuit performance critically rather than simply recording results.

Health and Safety Awareness.

Health and safety was consistently applied throughout all practical activities.

Safe working practices included:

- Following laboratory risk assessments.
- Conducting pre-operational equipment checks.
- Using ESD wrist straps when handling CMOS devices.
- Isolating power before modifying circuits.
- Checking component polarity before energising circuits.
- Maintaining a clean and organised workstation.
- Using correctly rated components and equipment.

These procedures reduced the risk of component damage, short circuits and injury.

Developing these habits is particularly important because professional engineers are expected to work safely while maintaining equipment reliability and protecting both personnel and assets.

2. Importance of Behaviours in Improving Personal Performance.

Time Planning and Management.

I approached the assignments systematically by dividing tasks into manageable stages.

For Assignment A, I completed:

1. Diode investigations.
2. Transistor investigations.
3. Operational amplifier investigations.

For Assignment B, I progressed through:

1. Combinational logic circuits.
2. Shift registers.
3. Ripple counters.

This structured approach allowed each activity to build upon previous knowledge.

However, I underestimated the amount of time required for hardware troubleshooting. The shift register prototype experienced switch bounce issues that required additional investigation and delayed testing.

In future projects I would:

- Create a detailed Gantt chart.
- Include contingency time for troubleshooting.
- Build prototypes earlier.
- Conduct regular progress reviews.

These improvements would reduce project risk and improve overall efficiency.

Communication and Literacy Skills.

Effective communication was essential throughout the assignments.

I demonstrated communication skills by:

- Following assignment instructions carefully.
- Interpreting component datasheets accurately.
- Producing structured technical reports.
- Using appropriate engineering terminology.
- Seeking clarification when necessary.

My reports included calculations, diagrams, simulation evidence and evaluations, helping communicate technical findings clearly.

One area for improvement is academic referencing. Future reports should include more consistent in-text citations to demonstrate greater research depth and academic rigour.

Self-Management and Professional Behaviours.

Several professional behaviours contributed to successful completion of the assignments.

Persistence.

When hardware faults occurred, particularly within the shift register prototype, I continued troubleshooting until the cause was identified and resolved.

Attention to Detail.

I carefully checked:

- Wiring connections.
- Component orientation.
- Measurement points.
- Logic state transitions.

This reduced assembly errors and improved reliability.

Self-Reflection.

I consistently compared theoretical, simulated and practical results, helping identify discrepancies and understand why differences occurred.

Professionalism.

Accurate record keeping and structured reporting demonstrated professional engineering practice throughout the unit.

One area for improvement would be adopting a more formal troubleshooting methodology rather than relying occasionally on trial-and-error approaches.

Application to Further Study and Employment.

The knowledge and skills developed during this unit provide a strong foundation for future engineering study.

Higher Education.

The analogue investigations support future study in:

- Electronic Engineering.
- Power Electronics.
- Analogue Circuit Design.

The digital investigations support further learning in:

- Microprocessors.
- Embedded Systems.
- FPGA Design.
- Digital System Architecture.

Employment.

The practical skills developed are directly transferable to engineering careers, including:

- Electronics Technician.
- Test Engineer.
- Maintenance Engineer.
- Design Engineer.

Proficiency with oscilloscopes, multimeters, simulations and fault-finding techniques is highly valued within industry.

The ability to analyse theory, simulation and practical results also reflects the quality assurance and diagnostic skills expected of professional engineers.

3. Conclusion and Evaluation.

What Went Well.

Several aspects of the project were highly successful.

- Correct Boolean expressions were derived and simplified using Karnaugh maps.
- Ripple voltage calculations closely matched simulation and practical results.
- The full-wave rectifier with Zener regulation maintained a stable 5.1V output.
- The common emitter amplifier achieved maximum undistorted gain.
- Frequency response testing confirmed expected gain roll-off.
- Multisim and CircuitVerse simulations consistently matched theoretical predictions.

- The combinational logic circuit operated correctly for all required input states.
- The shift register successfully transferred data between stages.
- The ripple counter counted correctly through all eight binary states.
- Health and safety procedures were followed consistently throughout.

These results demonstrate successful application of engineering principles and effective validation of circuit performance.

What Could Have Gone Better.

Although the investigations were successful, some limitations were identified.

- Component substitutions occasionally required recalculation of circuit values.
- Switch bounce caused instability within the shift register.
- Propagation delay produced glitches within the asynchronous counter.
- Some sections initially required additional editing and refinement.
- More comprehensive referencing could have strengthened academic quality.
- Troubleshooting required more time than originally planned.

These issues highlighted the importance of preparation, systematic fault-finding and realistic project planning.

What Would I Do Differently Next Time?

To improve future engineering projects, I would:

Technical Improvements

- Replace ripple counters with synchronous counters to eliminate propagation delay.
- Implement hardware debouncing circuits for clock inputs.
- Use PCB construction instead of breadboards where appropriate.
- Extend testing across a wider range of operating conditions.
- Record additional measurement data at multiple test points.

Project Management Improvements

- Produce a detailed Gantt chart before starting.
- Complete simulations before ordering components.
- Create a full bill of materials.

- Schedule dedicated troubleshooting time.
- Maintain a troubleshooting log throughout the project.

Documentation Improvements

- Include more annotated diagrams.
- Improve referencing throughout reports.
- Provide deeper analysis of discrepancies.
- Seek lecturer and peer feedback earlier.

Final Reflection.

Overall, Assignments A and B significantly enhanced my understanding of analogue and digital electronics while developing practical engineering skills and professional behaviours. The combination of theory, simulation and practical testing provided a systematic engineering approach that strengthened both technical competence and problem-solving ability. The experience has prepared me well for future study and employment within electronic engineering and has highlighted the importance of continuous improvement, safety awareness and reflective practice in achieving professional engineering standards.